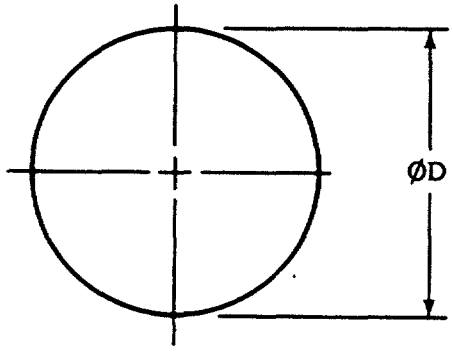
MILITARY SPECIFICATION SHEET
BALLS, BEARING, STEEL, CARBON, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the bearing described herein shall consist of this specification and the latest issue of DOD-B-82737.



NOTES:

1. All dimensions are in millimeters. Column headings in tolerance tables are in accordance with AFBMA-STD-10.
2. For tolerances by grade see Table II for individual balls and Table III for lots of balls. Tolerances are in micrometers (μm).
3. Grade 10 balls are preferred in most military applications.
4. Referenced government (or nongovernment) documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

FIGURE 1. BALLS, BEARING, STEEL, CARBON, METRIC.

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TABLE I. Dash numbers and dimensions.

ØD NOM	GRADE 100	GRADE 200	GRADE 500	GRADE 1000
	DASH NUMBERS			
1.5	10015	20015	50015	100015
2	10020	20020	50020	100020
2.5	10025	20025	50025	100025
3	10030	20030	50030	100030
3.5	10035	20035	50035	100035
4	10040	20040	50040	100040
4.5	10045	20045	50045	100045
5	10050	20050	50050	100050
5.5	10055	20055	50055	100055
6	10060	20060	50060	100060
6.5	10065	20065	50065	100065
7	10070	20070	50070	100070
7.5	10075	20075	50075	100075
8	10080	20080	50080	100080
8.5	10085	20085	50085	100085
9	10090	20090	50090	100090
10	100100	200100	500100	1000100
11	100110	200110	500110	1000110
11.5	100115	200115	500115	1000115
12	100120	200120	500120	1000120
13	100130	200130	500130	1000130

ØD NOM	GRADE 100	GRADE 200	GRADE 500	GRADE 1000
	DASH NUMBERS			
14	100140	200140	500140	1000140
15	100150	200150	500150	1000150
16	100160	200160	500160	1000160
17	100170	200170	500170	1000170
18	100180	200180	500180	1000180
19	100190	200190	500190	1000190
20	100200	200200	500200	1000200
21	100210	200210	500210	1000210
22	100220	200220	500220	1000220
23	100230	200230	500230	1000230
24	100240	200240	500240	1000240
25	100250	200250	500250	1000250
26	100260	200260	500260	1000260
28	100280	200280	500280	1000280
30	100300	200300	500300	1000300
32	100320	200320	500320	1000320
34	100340	200340	500340	1000340
35	100350	200350	500350	1000350
36	100360	200360	500360	1000360
38	100380	200380	500380	1000380

REQUIREMENTS:

1. Material: Steel, carbon conforming to the chemical composition of UNS G10080 thru UNS G10220 in accordance with ASTM A108 and as specified in procurement document.

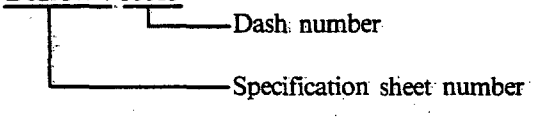
2. Hardness: Shall be 60 HRC or equivalent.

3. Surface Roughness: Tolerance limits as specified in Table II and in accordance with ANSI B46.1.

4. Material Density: Shall be 7.861 grams per cubic centimeter.

5. Part Number: Consists of the letter "D", the basic number of this specification sheet and a dash number taken from Table I.

Example: D82737/4-10015



D82737/4-10015 Indicates a grade 100 bearing ball of carbon steel with a basic diameter of 1.5 millimeters.

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TABLE II. Tolerances by grade for individual balls.

GRADE	ALLOWABLE BALL DIAMETER VARIATION[1/]	ALLOWABLE DEVIATION FROM SPHERICAL FORM[1/]	MAXIMUM SURFACE ROUGHNESS ARITHMETICAL AVERAGE[1/]
	V_D	W	
100	2.5	2.5	0.125
200	5	5	0.3
500	13	13	—[2/]
1000	25	25	—[2/]

[1/] Tolerances in micrometers (μ m)
[2/] Not applicable

TABLE III. Tolerances by grade for lots of balls.

GRADE	ALLOWABLE LOT DIAMETER VARIATION[1/]	BASIC DIAMETER TOLERANCE[1/]
100	5	± 12.5
200	10	± 25
500	25	± 50
1000	50	± 125

[1/] Tolerances in micrometers (μ m)

Custodians:
Army - AT
Air Force - 99

Review activities:
Navy - SH
Air Force - 84
DLA - IS

Preparing activity:
Navy - OS

Agent:
DLA - IS

(Project 3110-0737-04)

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